



Insertion Sort: Apni Sahi Jagah Banao!



1. Insertion Sort Kya Hai?

Maan lo tumhare UrbanPro wale Python students ko samajhna hai ki Insertion Sort kya hota hai. Isko real life se connect karte hain!

Tum gym jate ho, aur dumbbells ko unke rack mein wapas rakhna hai. Rack mein pehle se kuch weights line se rakhe hain (man lo 10kg, 15kg, 20kg). Ab tumhare hath mein **12kg** ka dumbbell hai.

Tum kya karoge? Tum rack mein 15kg aur 20kg ko thoda right mein shift karoge aur **12kg** wale dumbbell ko uski **sahi jagah (insert)** kar doge! Yahi same logic Insertion sort ka hai.



2. Step-by-Step Logic

Hum array ko do hisso mein batte hain: ek jo **Sorted (arrange ho chuka hai)** aur dusra **Unsorted (jo bacha hai)**.

Numbers: 64, 25, 32, 20

Step 1: Pehla number (64) toh akela hai, usko sorted maan lo. Ab bari aayi **25** ki. Ise hum '**Key**' (🔑) bana lenge.



Key = 25. Kya 64, 25 se bada hai? Haa! Toh 64 ko ek kadam right mein (aage) **shift** kar do, aur 25 ko 64 ki jagah daal do (insert kar do).



Step 2: Ab humare array ke shuruwat ke do numbers (25, 64) sorted hain. Ab bari aayi **32** ki.

Key = 32. Hum 32 ke left mein check karenge. Kya 64 bada hai 32 se? Haa, toh 64 ko right shift karo. Kya 25 bada hai 32 se? Nahi! Toh 32 ko 25 ke baad insert kar do.



Isi tarah hum ek-ek number uthate hain (Key banate hain) aur use left side ke sorted numbers ke beech uski sahi jagah par daal dete hain.



3. Python Code for Insertion Sort

Chalo UrbanPro students ke liye ekdum clear code likhte hain:

```
def insertion_sort(A):
    n = len(A)
    # Loop 1 se shuru hoga, kyunki 0 index wala already sorted
    maante hain
    for i in range(1, n):
        key = A[i] # Ye hamara target dumbbell (number) hai
        j = i - 1 # Left side ke sorted numbers se compare
        karenge

        # Jab tak j >= 0 hai aur left wala number humare 'key'
        se bada hai
        while j >= 0 and A[j] > key:
            A[j + 1] = A[j] # Bade number ko right mein shift
```

```

karo
        j -= 1          # Ek kadam peeche aao check karne
ke liye

        # Jab shift karna khatam ho jaye, toh Key ko uski sahi
jagah rakh do
        A[j + 1] = key

# Test Karte Hain!
arr = [64, 25, 32, 20, 15]
insertion_sort(arr)
print("Sorted Array:", arr)

```

Important Points to Remember!

- **The "Key":** Har bar ek naya element **Key** banta hai. Hum key ko compare karte hain uske left wale sabhi elements se.
- **Shift, NOT Swap:** Isme hum Bubble ya Selection sort ki tarah numbers ko Swap (jagah badalna) nahi karte, balki numbers ko aage **Shift (khiskana)** karte hain.
- **Time Complexity:** Iski Worst Case Time Complexity bhi **$O(n^2)$** hoti hai.
- **Best Use:** Yeh algorithm tab bahut fast kaam karta hai jab array already aadha sorted ho!

Keep coding, keep growing! 